

A quadratic limitation of disjoint linear moment corrections

Research note, 30 September 2026

Status. Complete elementary proof. This is a limitation of the new construction method, not a lower bound for arbitrary fair dice. It explains why reducing the number of nodes in each private correction set cannot turn that method into a linear-size exceptional die.

Proposition 1. *Let $m \geq 2$, and let B be a nonzero linear functional on the polynomials of degree at most $m - 1$. Suppose that, for each $j \in [m]$, it has a signed atomic representation*

$$B(a) = \sum_{q \in Q_j} \delta_{j,q} a(t_q) \quad (\deg a \leq m - 1),$$

where all t_q are distinct, the sets $Q_1, \dots, Q_m$ are pairwise disjoint, and all displayed weights are nonzero. Then

$$|Q_i| + |Q_j| \geq m + 1 \quad (i \neq j), \quad \sum_{j=1}^m |Q_j| \geq \frac{m(m+1)}{2}.$$

More precisely, minimizing the integer sum under these pair inequalities requires at least

$$\begin{cases} m(m+1)/2, & m \text{ odd}, \\ m(m+2)/2 - 1, & m \text{ even}. \end{cases}$$

Proof. Subtract the representations for i and j . This gives a nonzero signed measure with support $Q_i \cup Q_j$ whose moments of orders $0, \dots, m - 1$ vanish. A signed measure supported on $s \leq m$ distinct points and with these vanishing moments must be zero: apply the first s moment equations, whose coefficient matrix is an invertible Vandermonde matrix. Since the supports are disjoint and all weights are nonzero, the difference measure is nonzero. Thus $|Q_i| + |Q_j| \geq m + 1$. Summing over unordered pairs gives

$$(m-1) \sum_j |Q_j| \geq \binom{m}{2} (m+1).$$

For the sharper integer statement, write $a_1 \leq \dots \leq a_m$ for the support sizes. Only $a_1 + a_2 \geq m + 1$ matters for minimization. If $m = 2r + 1$, taking all $a_j = r + 1$ attains the displayed bound. If $m = 2r$, either $a_1 \geq r + 1$, giving total at least $m(r+1)$, or $a_1 \leq r$ and $a_j \geq 2r+1-a_1$ for $j \geq 2$. The latter sum is minimized at $a_1 = r$, and equals $r + (m-1)(r+1) = m(m+2)/2 - 1$. $\square$

Application to the fair-dice construction. The construction in `size_bounds/polynomial_except` uses $m = n - 1$ old dice and private node sets Q_j for their density corrections. After absorbing the factor $1/K$ into the signed weights, its common correction functional is exactly of the form above. Any unused nodes only increase the total distinguished-die size. Thus a nonzero correction forces $K = \Omega(m^2)$ within this method.

If $B = 0$, the uncorrected rational distinguished nodes already form a scalar rational equal-weight interval quadrature of degree m . The companion two-adic theorem then forces $K \geq 2^{\lfloor m/2 \rfloor}$. In particular, asymptotically polynomial versions of this disjoint-support construction necessarily use a nonzero correction, and the quadratic limitation applies.

The remaining exceptional-die problem. Let $g(n)$ be the minimum number of faces of a prescribed die in an n -die permutation-fair collection, with the other dice unrestricted. The results obtained in this research session give

$$n - 1 \leq g(n) \leq Cn^2, \quad g(2) = 1, \quad g(3) = 2, \quad g(4) = 3.$$

The proposition does not decide whether $g(n)$ has linear or quadratic order. A construction with $o(n^2)$ distinguished faces would need a mechanism beyond disjoint private supports representing one common nonzero correction functional in this way.